

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

11th September 2018

Subject ST8 – General Insurance: Pricing
Time allowed: Three Hours (14.45* – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Outline the balance sheet of a general insurance company and its major components. [3]
- Q. 2)** You are the pricing analyst with a general insurance company selling only motor insurance. As a part of the pricing exercise, you are modeling claims severity.
- i)** Write down why you would use spatial smoothing for the Pin Code as a rating factor. (1)
- ii)** State along with justification what form of spatial smoothing, distance-based or adjacency-based method, would be more suitable for theft claims in Motor insurance. (2)
- [3]
- Q. 3)** Define Parameter Uncertainty. List the sources through which parameter uncertainty could arise in the context of general insurance business. [4]
- Q. 4)** You are the pricing actuary of a general insurance company which predominantly underwrites property insurance for many years. The company has gathered credible and reliable premium and claims data over the years for this class of business. You are in the process of revising the premium rates for this business. The revised rates shall be applicable for next three years.
- State the adjustments, both prospective and retrospective, that you shall be doing in the data to calculate the premium rates. [4]
- Q. 5)** The CEO of your company has received a report on claims frequencies for its motor insurance business. The report shows that the policyholders owning high-end vehicles have higher claim frequency as compared to the policyholders owning small size vehicles. The CEO also knows that the Company is charging lower premium rates to high-end vehicles as compared to small size vehicles. Premium rates are applied on the value of the car to arrive at the premium.
- The CEO would like to understand the reasons for charging lower premium rates to higher-end vehicles despite having higher claims frequency.
- Discuss the investigations that should be done to reply to the comments of the CEO. [5]
- Q. 6)** A large general insurance company sells only personal lines insurance through individual agents. The company primarily sells only two products:
- Motor Insurance covering damage to the vehicle and third party liability
 - Home insurance covering structure and contents
- The company has significant market share in both classes of business. However, there is drop in market share for motor policy in the last few years. Management is concerned that one of the reasons could be expense cross-subsidies between the two products which are causing the motor policies to be more expensive than the competitors.
- Outline the analyses that you would do to help determine the causes of drop in market share. [6]
- Q. 7)** You are the pricing analyst of an auto insurance company who is building a generalized linear model (GLM) to predict the frequency of own damage claims for a book of private car vehicles. Your manager has suggested you to have a relook at the risk classification system for vehicles as well as for drivers. He also suggested you to think of new rating factors to include in the system for better segmentation.

You discussed the same with your actuarial colleagues and someone suggested to add “number of hours a driver sleeps each night” as a rating variable.

- i) Discuss the comments in favour of and against this recommendation. (4)
- ii) One of your colleague also suggested to change the exposure base from vehicle years to hours driven. (3)

Evaluate the use of hours as a good exposure base. [7]

- Q. 8)** You work for an insurance company underwriting all lines of business in a developing country. In the country, the State provides its citizens who are below a certain level of income, with a personal accident cover of Rs. 1,00,000. The premium for the cover is borne completely by the State. The cover is provided on the basis the census data i.e. the total premium is calculated as premium per life times the total number of lives covered. The State is running the scheme for the last 10 years.

During the initial 8 years, the State had opted for insurance model to provide the cover. Under this model, all the insurers operating in the country were invited for bidding and the policy is awarded to the insurer with the lowest quote.

For the last 2 years, the State had decided to opt for the trust model, wherein the State had formed a trust which shall settle all claims under the policy using the funding done by the State for this purpose.

Citing operational difficulties, the State has decided to go back to the insurance model.

Your Company has never placed a bid for this business in the past. The Chief Marketing Officer is looking to underwrite this business depending on the profitability of the scheme. Your Company is seeking the historic data from the State to price this scheme.

- i) Discuss the underlying concerns in the data that you have received from the State for the purpose of pricing this scheme. (4)
 - ii) Discuss the checks that you shall perform on the data to establish the reasonableness of the data received. (2)
 - iii) List the actions that could be taken to mitigate the effects of any potential inadequacies or inaccuracies in the data provided by the State. (2)
- [8]

- Q. 9)** A general insurance company provides a fixed home insurance cover of amount “M”. The expected claims distribution of the policy is as given below:

Claim Payment (in Rs.)	Probability
0	95%
0.03M	2%
0.14M	1.5%
0.57M	1%
M	0.5%

- i) Calculate the pure premium as a proportion of Sum Insured, assuming all householders purchase full coverage. State any assumption that you make. (2)

ii) The insurance company has noted that a subset of homeowners purchase only partial coverage. In one example, the insured purchased a cover of Rs.10 Lacs to cover his home when the Sum Insured should have been Rs. 35 Lacs. Illustrate how in this case, the use of fixed rate per sum insured on Rs. 10 Lacs is not equitable for the insurer. (4)

iii) Discuss and justify the steps that can be taken by the Company to reduce the inequity from part **(ii)** above (4)
[10]

Q. 10) In a developing country, Insolvency and Bankruptcy Code was recently passed which allows the creditors of a defaulting company to take the company to the court and ask for a resolution against the default amount. This can be achieved by either change of management or change of business operations through a resolution plan ratified by all the creditors or liquidation in case the resolution plan fails to take off for some reasons.

The business is revalued and bid under the proposed resolution plan by a willing bidder. After winning the bidding process the successful resolution plan is submitted to the committee of creditors where a majority approval is required for the plan to be finally implemented.

The so-called interim resolution professionals are responsible for coordinating and facilitating the resolution plan and they take over the management of the business operations during the resolution process.

i) State the possible risks faced by the resolution professionals under this code while executing the resolution plan. (3)

ii) State the insurance coverages which may be sought by such resolution professionals and describe the features/characteristics of these coverages. (8)
[11]

Q. 11) i) State the two broad categories under which a catastrophic model can be classified on the basis of level of detail in the Inventory Module. (1)

ii) The new Crop Insurance scheme recently introduced by the government of a developing country is on an area based approach wherein losses including catastrophic losses are assessed for settlement at an aggregate level or a larger area level instead of the individual farmland level.

Under this mechanism, the farmers carry the basis risk in that the claim settlement amount might differ from the actual losses suffered.

a) State the sources of basis risk in crop insurance. (2)

b) For losses due to natural perils, the scheme makes use of catastrophe models for settlement of claims arising due to natural perils. State which category of Catastrophic models mentioned under part **(i)** of this question above can help to eliminate the basis risk above and how. (2)

c) State the challenges in implementing the category of catastrophe model suggested under part **(ii) (b)** above and also the reasons for the same in respect of this crop business. (4)

- d)** Briefly explain how the challenges mentioned above under part **(ii) (c)** can be overcome. (3)
[12]

- Q. 12) i)** With regard to the use of data in pricing, state the various sources of data error. Also briefly explain the consequences of each one of them on the financials of an insurance company. (4)
- ii)** A pricing exercise is being undertaken by a reinsurer for an adverse development cover to be provided to a reinsured client. State with reasons the sources of data errors, as mentioned under part **(i)** above, prevailing in the existing system of the reinsured client which can distort the results significantly. (2)
- iii)** In a large general insurance company, during one of the regular system audits, it has been found that a re-open claim is assigned a completely different claim number than the original one. In response to the audit query a student actuary responds by saying that it has been the legacy practice in the organisation.
- a)** As a senior actuary, prepare a response to the student stating the reasons and consequences of having followed this particular practice. (3)
- b)** State the immediate consequences and challenges if the existing practice of treating the re-open claims is now changed to a more appropriate practice. (3)
[12]

- Q. 13) i) a)** State the assumptions made under the Buhlmann-Straub Formula. (3)
- b)** State the other credibility model which the Buhlmann-Straub model is similar to and also state the suitable reasons for the similarity. (2)
- ii)** Under the Government Health Insurance Scheme, there are two classes of policyholders who are to be covered by the scheme for a Sum Insured of 5 Lacs.

Assume that the annual historical claims profile of the two classes of policyholders, Class A and Class B, are known and are as follows:

A randomly selected family is equally likely to have come from Class A or Class B.

Annual Claim Count Distribution of Class A type of Policyholders

Number of Claims made by a family	Family Count	Total No. of claims
0	4000	0
1	2000	2000
2	1000	2000
Total	7000	4000

Annual Claim Size Distribution of Class A type of Policyholders

Claim Amount made by a family	Count
1000	3500
50000	400
75000	100
Total	4000

Annual Claim Count Distribution of Class B type of Policyholders

Number of Claims made by a family	Family Count	Total No. of claims
0	2000	0
1	3000	3000
2	2000	4000
Total	7000	7000

Annual Claim Size Distribution of Class B type of Policyholders

Claim Amount made by a family	Count
1000	6400
50000	500
75000	100
Total	7000

The variance of the losses/claims for a family (Class of policyholders is not fixed) in a year is 2300000000.

A randomly selected family is observed to have annual losses of Rs. 2000 in the previous year.

Determine the Buhlmann credibility premium which should be charged to this family for next year under the new Health Scheme. Ignore all the expense margins for loading for simplicity.

State all the other assumptions made.

(7)

- iii)** State with reasons any particular assumption(s) under Buhlmann-Straub Model which may possibly be violated under the example in part **(ii)** above.

(3)

[15]
